

Local Projection and VARs May Estimate the Same Impulse Responses

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1 Introduction

Impulse Response Functions (IRFs) are fundamental tools in macroeconometrics to trace the dynamic effects of a shock to one variable on the rest of the system. They are commonly used to understand the propagation mechanisms of economic shocks over time.

In applied macroeconometrics, the most widely used empirical models for estimating IRFs have traditionally been Structural Vector Autoregressive (SVAR) models. Over the past two decades, however, starting with the influential work of Jordà (2005), *Local Projections* (LPs) have emerged as a popular alternative method for estimating impulse responses.

A key contribution in this area is provided by Plagborg-Møller and Wolf (2021) in their paper "*Local Projections and VAR Estimate the Same Impulse Responses*". The authors show that LPs and VARs are not conceptually distinct procedures. In fact, they demonstrate that structural identification methods commonly used in the VAR literature—such as short-run, long-run, and sign restrictions—can be equivalently applied within the LP framework.

These are the core concepts on which the present analysis builds, although the original paper also contains additional results and technical refinements that are not discussed here.

They further note that, in real-world applications where it is not feasible to include an infinite number of lags, the impulse responses produced by VARs and LPs coincide up to a forecast horizon equal to p . In this sense, LPs and VARs are simply two linear projection techniques.

Crucially, structural identification defines how a shock is identified, not how the response is estimated. Both methods aim to recover the same best linear approximation to the true impulse response function, which, in practice, may be nonlinear. This result emphasizes that the identification strategy is conceptually separate from the choice of estimation method.

I found their contribution particularly insightful because (i) their results do not rely on strong parametric assumptions, and (ii) they extend long-run and sign-restriction identification strategies to the local projection framework.

In Section 3.1 of their paper, Plagborg-Møller and Wolf (2021) present three illustrative examples that demonstrate the mathematical logic by which VARs and LPs estimate the same impulse response functions. These examples correspond to standard identification strategies widely used in the macroeconometrics literature:

- **Example 1 — Recursive Identification:** Christiano, Eichenbaum, and Evans (2005)
- **Example 2 — Long-run Identification:** Blanchard and Quah (1989)
- **Example 3 — Sign Identification:** Uhlig (2005)

In this work, I use data from the past 40 years of the U.S. economy to replicate the structural models from the three papers cited above.

The sample period is 1984:Q1–2024:Q4.

While I am aware that the structure of the economy and the relationships between macroeconomic variables have evolved significantly over time, according to the results of Plagborg-Møller and Wolf (2021), if their assumptions hold, the equivalence between VAR-based and LP-based impulse responses should still be valid.

In each example, I will use the same set of variables as in the original studies and, where possible, apply the same data transformations. However, my primary concern will be to ensure that the key assumptions outlined in Plagborg-Møller and Wolf (2021) are satisfied.

All data are from the quarterly macroeconomic database provided by McCracken and Ng (2021), officially known as FRED-QD (Federal Reserve Economic Data – Quarterly Database), unless otherwise stated.

The sole objective of this work is to assess whether, under the assumptions laid out by Plagborg-Møller and Wolf (2021), the impulse response functions (IRFs) estimated via VARs and Local Projections coincide. Issues related to inference and economic interpretation lie beyond the scope of this analysis.

2 Conditions for Equivalence Between LPs and VARs

The notations, assumptions, and propositions presented below are taken directly from Plagborg-Møller and Wolf (2021), and are reproduced here for clarity.

The data are defined as:

$$w_t = \begin{pmatrix} r'_t \\ x_t \\ y_t \\ q'_t \end{pmatrix}$$

where:

- r_t is an $n_r \times 1$ vector of control variables (time series),
- q_t is an $n_q \times 1$ vector of control variables (time series),
- x_t is the source of the shock (scalar time series),
- y_t is the response variable (scalar time series).

Assumption 1

The data $\{w_t\}$ are covariance stationary and purely nondeterministic, with absolutely summable Wold decomposition coefficients.

In particular, no assumptions about the underlying causal structure of the economy are made.

Assumption 2

$\{w_t\}$ is a jointly Gaussian vector time series.

Gaussianity is assumed for ease of notation, enabling the use of conditional expectations. Dropping this assumption does not affect the results, as they continue to hold under linear projections.

Local Projection

Consider, for each $h = 0, 1, 2, \dots$, the following linear projection:

$$y_{t+h} = \mu_h + \beta_h x_t + \gamma_h' r_t + \sum_{\ell=1}^{\infty} \delta_{h,\ell}' w_{t-\ell} + \xi_{h,t} \quad (1)$$

Here, $\xi_{h,t}$ denotes the projection residual, and $\mu_h, \beta_h, \gamma_h, \delta_{h,1}, \delta_{h,2}, \dots$ are the corresponding projection coefficients.

IRF The LP impulse response function of y_t with respect to x_t is defined by the sequence $\{\beta_h\}_{h \geq 0}$ in equation (1).

This expression defines the LP impulse response estimand at horizon h as:

$$\beta_h = \mathbb{E}(y_{t+h} \mid x_t = 1, r_t, \{w_\tau\}_{\tau < t}) - \mathbb{E}(y_{t+h} \mid x_t = 0, r_t, \{w_\tau\}_{\tau < t}) \quad (2)$$

Note that equation (1) includes the current value of r_t , but not q_t . We do not assume that x_t is a predetermined shock. Still, the formulation is general enough to cover most empirical applications of local projections.

VAR

We now consider the multivariate linear VAR(∞) representation:

$$w_t = c + \sum_{\ell=1}^{\infty} A_\ell w_{t-\ell} + u_t, \quad (3)$$

where u_t is the projection residual, defined as $u_t = w_t - \mathbb{E}(w_t \mid \{w_\tau\}_{\tau < t})$, and $c, A_1, A_2, \dots$ are the projection coefficients.

Let $\Sigma_u = \mathbb{E}(u_t u_t')$ be the residual covariance matrix, and let its Cholesky decomposition be $\Sigma_u = BB'$, where B is lower triangular with positive diagonal elements.

This leads to the recursive SVAR representation:

$$A(L)w_t = c + B\eta_t,$$

where $A(L) = I - \sum_{\ell=1}^{\infty} A_\ell L^\ell$ and $\eta_t = B^{-1}u_t$.

Define the lag polynomial $C(L) = \sum_{\ell=0}^{\infty} C_\ell L^\ell = A(L)^{-1}$.

IRF The VAR impulse response function of y_t to a shock in x_t is:

$$\theta_h = C_{h, n_r+2, \cdot} B_{\cdot, n_r+1},$$

where C_h is the h -th coefficient matrix from $C(L)$, and B is as above. Here, $C_{h,i,\cdot}$ denotes the i -th row of C_h , and $B_{\cdot,j}$ is the j -th column of B .

The ordering of r_t and q_t follows common practice in applied macroeconometrics.

Proposition 1: *Under Assumptions 1 and 2, the LP and VAR impulse response functions are equal, up to a constant of proportionality:*

$$\theta_h = \sqrt{\mathbb{E}(\tilde{x}_t^2)} \times \beta_h \quad \text{for all } h = 0, 1, 2, \dots,$$

where

$$\tilde{x}_t \equiv x_t - \mathbb{E}(x_t \mid r_t, \{w_\tau\}_{\tau < t}).$$

The presence of the constant of proportionality arises because the implicit LP innovation $\tilde{x}_t$, after controlling for the other right-hand side variables, does not have unit variance.

If we rescale $\tilde{x}_t$ to have variance one, or if we consider relative impulse responses θ_h/θ_0 , then the LP and VAR impulse response functions coincide exactly.

Identification Under Invertibility

Standard SVAR analysis assumes (partial) invertibility, that is, the ability to recover the structural shock of interest, $\varepsilon_{1,t}$, as a function of only current and past macro aggregates:

$$\varepsilon_{1,t} \in \text{span}(\{w_\tau\}_{-\infty < \tau \leq t}). \quad (4)$$

A given SVAR identification scheme then identifies the candidate structural shock as a particular linear combination of the Wold forecast errors:

$$\tilde{\varepsilon}_{1,t} \equiv b' u_t, \quad (5)$$

where the chosen identification scheme determines the vector b as a function of the reduced-form VAR parameters $(A(L), \Sigma_u)$, or equivalently, of the Wold decomposition parameters $(C(L), \Sigma_u)$.

Under invertibility, there must exist a vector b such that $\tilde{\varepsilon}_{1,t} = \varepsilon_{1,t}$, implying that SVAR identification can, in principle, succeed (Fernandez-Villaverde (2007), Rubio-Ramirez (2007), Wolf(2020)).

In practice, invertibility corresponds to checking whether the estimated VAR is invertible in the Wold sense. This involves estimating the VAR model and verifying that all eigenvalues λ of the associated companion matrix lie inside the unit circle, that is, $|\lambda| < 1$. This condition ensures the stability and invertibility of the VAR representation, which is a necessary condition for structural identification to be valid.

Assumptions Underlying the Empirical Implementation

In this work, the main criteria for the validity of the results are the use of stationary time series and the requirement that all eigenvalues λ of the VAR-associated companion matrix lie strictly inside the unit circle.

As for the identification of structural shocks, I assume that—despite structural changes in the economy over time—some key relationships have remained stable. Therefore, I adopt the identification strategies and the lag p proposed in the original reference papers.

3 Recursive Identification: Christiano, Eichenbaum, and Evans (2005)

Variables Used in CEE (2005)

The set of macroeconomic variables used in the VAR by Christiano, Eichenbaum, and Evans (2005) is structured as follows:

$$Y_t = [Y_{1t} \quad R_t \quad Y_{2t}]'$$

- Y_{1t} includes variables that do not respond contemporaneously to the monetary policy shock:
 - real gross domestic product (GDP)
 - real consumption
 - the GDP deflator
 - real investment
 - the real wage
 - labor productivity
- R_t is the federal funds rate, interpreted as the monetary policy instrument.
- Y_{2t} includes variables that are allowed to respond contemporaneously:
 - real profits
 - the growth rate of M2

In this example all the variables are available from the FRED-QD (Federal Reserve Economic Data – Quarterly Database), except for the Real Profits. I build this variables by taking the Corporate profits before tax time series (*A053RC1Q027SBEA* in the FRED website) and dividing it by the GDP Deflator.

I applied first-differencing to all variables. After this transformation, all series were found to be stationary except for real GDP and the GDP deflator, which remained non-stationary according to ADF tests.

VAR Approach

Christiano, Eichenbaum, and Evans (2005) estimate a VAR model to analyze the effects of monetary policy, adopting the following specification:

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + A_3 Y_{t-3} + A_4 Y_{t-4} + C \eta_t, \quad (6)$$

where:

- Y_t is a vector of macroeconomic variables ordered according to a causal structure (temporal ordering),
- A_i are matrices of autoregressive coefficients,
- C is an $n \times n$ lower triangular matrix with diagonal elements normalized to 1,
- the vector of structural shocks η_t is assumed to have zero mean, to be serially uncorrelated, and to have a diagonal variance-covariance matrix:

$$\mathbb{E}[\eta_t \eta_t'] = \Sigma_\eta, \quad \text{with } \Sigma_\eta \text{ diagonal.}$$

After estimating the VAR model using the same variable ordering as in the original paper, I computed the matrix C via Cholesky decomposition and normalized it to have ones on the diagonal.

The impulse response functions (IRFs) were estimated using the standard formula:

$$IRF_{r|c,s} = \sum_{h=0}^{s-1} \epsilon_r' \Phi_h C \epsilon_c,$$

where Φ_h denotes the h -step ahead moving average coefficient matrices, and ϵ_r , ϵ_c are selection vectors corresponding to the response and shock variables, respectively.

Since all variables entered the model in first differences, I used *cumulative* IRFs to recover the effect on the level of the variables. The cumulative IRFs are defined as:

$$C_s = \Phi_s = \sum_{h=0}^{s-1} \Phi_h,$$

or equivalently, using the companion matrix representation:

$$C_s = \Phi_s = \sum_{h=0}^{s-1} A^h.$$

LP Approach

The local projection (LP) method estimates, for each horizon h , the following regression:

$$y_{t+h} = \mu_h + \beta_h x_t + \gamma_h' r_t + \sum_{\ell=1}^p \delta_{h,\ell}' w_{t-\ell} + \xi_{h,t},$$

where the components map to the structure of the CEE (2005) VAR as follows:

- **Response variable** (y_{t+h}): one of the nine elements of the vector

$$Y_t = [Y_{1t}' \quad R_t \quad Y_{2t}']',$$

considered one at a time.

- **Shock regressor** (x_t): the structural monetary policy shock $\eta_{MP,t}$, extracted from the VAR residuals via $x_t = C^{-1}u_t$.
- **Contemporaneous controls** (r_t): the six “predetermined” variables in Y_{1t} — real GDP, real consumption, GDP deflator, real investment, real wage, and labor productivity — which are not allowed to respond on impact.
- **Additional controls** (q_t): the two contemporaneous variables in Y_{2t} — real profits and M2 growth — which are allowed to respond on impact.
- **Lagged controls** ($w_{t-\ell}$): the full vector $w_t = (r_t', x_t, y_t, q_t')'$ lagged by $\ell = 1, \dots, p$ periods.

The coefficient β_h captures the impulse response of the selected variable at horizon h to a one-standard-deviation monetary policy shock, conditional on r_t and the full history of $w_{t-\ell}$.

Since all variables entered the model in first differences, I compute *cumulative* LP impulse responses to recover the effects on the level of the variables. This transformation ensures that LP-based responses are directly comparable to the level responses recovered from the VAR model.

VAR vs LP Cumulative IRFs: Christiano, Eichenbaum, and Evans (2005)

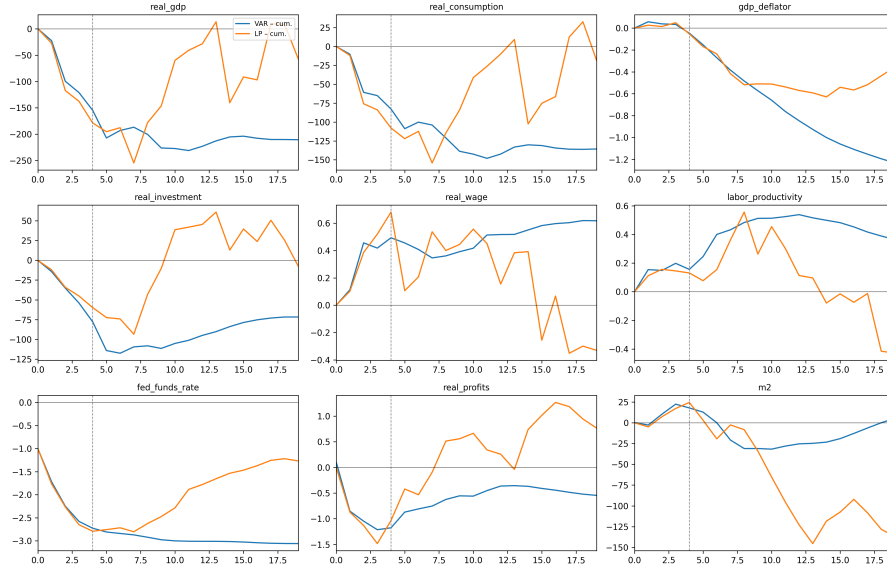


Figure 1: The dashed vertical line indicates the lag order p used in estimation.

The results are consistent with the findings of Plagborg-Møller and Wolf (2021). The IRFs estimated via VAR and local projections exhibit the same dynamic behavior up to the specified lag order p . In some cases, the responses are remarkably close across the two methods. Beyond horizon p , however, the dynamic patterns tend to be different, as expected.

4 Long-run Identification: Blanchard and Quah (1989)

Blanchard and Quah (1989) propose a bivariate VAR model to identify supply and demand shocks, focusing on two macroeconomic variables: the unemployment rate and the level of output, specifically real Gross National Product (GNP). Their key identifying assumption is that demand shocks have no long-run effect on output.

I follow the original paper in using $p = 8$ lags.

Variables Used in Blanchard and Quah (1989)

In this work, I follow their methodological framework but use Gross Domestic Product (GDP) in place of GNP. To maintain consistency with their approach, I transform my GDP series analogously to how they transformed GNP, applying the same differencing and other relevant transformations. The unemployment rate is left in levels, just as in their original study.

However, unlike Blanchard and Quah, I do not perform the demeaning and aggregation steps they applied to their data. Instead, I work directly with quarterly data as provided, which simplifies the data preparation while still allowing for a valid replication of their identification strategy.

VAR Approach

The identification is achieved through direct decomposition of the long-run variance of the reduced-form system.

Let $A_1, \dots, A_p$ be the coefficient matrices estimated from the reduced-form VAR. I begin by computing the sum of these matrices to obtain:

$$A_\Sigma = A_1 + A_2 + \dots + A_p.$$

From this, the matrix $A(1)$, representing the long run equilibrium, is given by:

$$A(1) = I - A_\Sigma.$$

I then compute the inverse of $A(1)$:

$$A(1)^{-1}.$$

Given the residual covariance matrix Ω from the reduced-form VAR, I construct the long-run variance matrix of the observable variables as:

$$V = A(1)^{-1} \Omega A(1)^{-1'}.$$

This matrix represents the unconditional long-run variance of the reduced-form process. To identify structural shocks, I perform a Cholesky decomposition of V :

$$V = \Phi(1)^* \Phi(1)^{*'},$$

where $\Phi(1)^*$ is a lower-triangular matrix. This decomposition implicitly imposes the identifying restriction that the second structural shock (interpreted as demand) has no long-run impact on the first variable (GDP), i.e., $[\Phi(1)^*]_{1,2} = 0$.

Next, I recover the long-run structural impact matrix B as:

$$B = A(1) \cdot \Phi(1)^*.$$

This matrix encodes the cumulative long-run effect of structural shocks on the system. To obtain the contemporaneous impact matrix C , I solve:

$$C(1)^* = A(1)^{-1} B.$$

Finally, structural impulse response functions (IRFs) at each horizon h are computed by post-multiplying the reduced-form response matrices Φ_h by the identified C :

$$\Psi_h^* = \Phi_h \cdot C(1)^*.$$

Since GDP enters the VAR in first differences, I compute the level response by the cumulative IRFs.

For unemployment, which is already expressed in levels, standard (non-cumulative) IRFs are used.

LP Approach

Blanchard and Quah (1989) identify structural shocks by imposing that *demand shocks have no long-run effect on GDP*. Wolf (2021) demonstrates that this same restriction can be implemented using a Local Projection (LP) approach based on regressions over long horizons.

Step 1: Identify the supply shock direction

The first step is to isolate the supply shock, which is the component of the data responsible for permanent changes in output. To do this, we estimate the following long-difference projection:

$$\Delta \log(gdp_{t+H}) - \Delta \log(gdp_{t-1}) = \tilde{\mu}_H + \tilde{\beta}'_H w_t + \sum_{\ell=1}^p \tilde{\delta}'_{H,\ell} w_{t-\ell} + \tilde{\xi}_{H,t},$$

where:

- $w_t = (\Delta \log gdp_t, unr_t)'$ is the vector of observed variables,
- H is a sufficiently large horizon (e.g., 40 quarters in this case),
- p is the number of lags included to control for dynamics.

The coefficient vector $\tilde{\beta}_H$ captures the linear combination of w_t that best explains the long-run variation in output.

To obtain a properly scaled direction, we normalize $\tilde{\beta}_H$:

$$b = \frac{\tilde{\beta}_H}{\|\tilde{\beta}_H\|}.$$

This normalized vector b identifies the *supply shock* direction — that is, the component of w_t associated with permanent effects on output.

Step 2: Identify the demand shock direction

Once the supply shock direction b is identified, the *demand shock* direction $\tilde{b}$ must be orthogonal to b in the sense of forecastable variation. Formally, we impose:

$$\tilde{b}' \text{Var}(w_t \mid \text{past}) b = 0,$$

meaning that $\tilde{b}$ is uncorrelated with b after controlling for the past history of w_t .

Since we are working with only two variables, the condition that $\tilde{b}$ is orthogonal to b in forecastable space simplifies to a geometric orthogonality in two-dimensional Euclidean space.

If we write the supply shock direction as a vector

$$b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix},$$

then any vector $\tilde{b}$ that is orthogonal to b must satisfy

$$b' \tilde{b} = b_1 \tilde{b}_1 + b_2 \tilde{b}_2 = 0.$$

A simple choice that satisfies this is

$$\tilde{b} = \begin{bmatrix} -b_2 \\ b_1 \end{bmatrix}.$$

This works because the dot product of b and $\tilde{b}$ is

$$b' \tilde{b} = b_1(-b_2) + b_2 b_1 = 0,$$

which proves orthogonality.

Thus, in two dimensions, this operation corresponds to a 90-degree rotation of the vector b , ensuring that $\tilde{b}$ is perpendicular to b . This guarantees that the demand shock direction $\tilde{b}$ is uncorrelated with the supply shock direction b , fulfilling the identifying restriction posed by Wolf (2021).

Step 3: Estimate impulse responses

With the directions b and $\tilde{b}$ representing the supply and demand shocks respectively, we estimate impulse responses using Local Projections. For each horizon h , we run the regression:

$$y_{t+h} = \mu_h + \beta_h(b'w_t) + \sum_{\ell=1}^p \delta_{\ell,h}w_{t-\ell} + \xi_{h,t}.$$

Here:

- $y_{t+h} - y_t$ is the cumulative change in the variable of interest,
- $b'w_t$ (or $\tilde{b}'w_t$) is the structural shock regressor at time t ,
- The lagged terms $w_{t-\ell}$ control for the past dynamics,
- β_h estimates the impulse response at horizon h .

This regression is performed twice: once using $b'w_t$ to obtain the supply shock IRFs, and once using $\tilde{b}'w_t$ for the demand shock IRFs. This approach allows us to recover structural impulse responses without estimating a full VAR system.

Remark Since GDP is modeled in first differences, *cumulative* impulse responses are computed to interpret effects on levels, while for unemployment rate (already in levels) the standard IRFs are directly used.

In this case, it is useful to visualize the impulse response functions (IRFs) from the two methods in separate graphs, as the difference in scale is clearly noticeable. Figure 4, instead, shows the IRFs scaled by the variance of the LP shock, allowing for a more direct comparison between the methods.

VAR Cumulative IRF: Blanchard and Quah (1989)

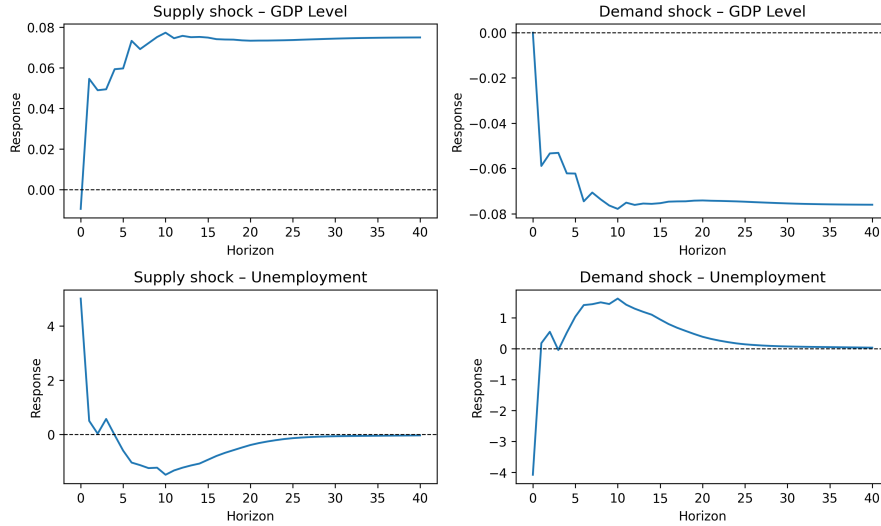


Figure 2: Only the GDP response is cumulative; Unemployment rate is in levels.

LPs Cumulative IRF: Blanchard and Quah (1989)

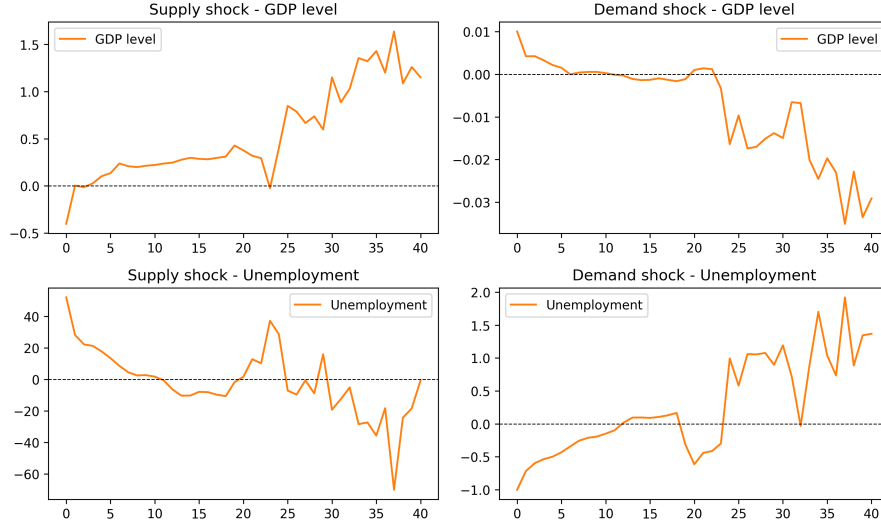


Figure 3: Only the GDP response is cumulative; Unemployment rate is in levels.

VAR vs LP Scaled Cumulative IRF: Blanchard and Quah (1989)

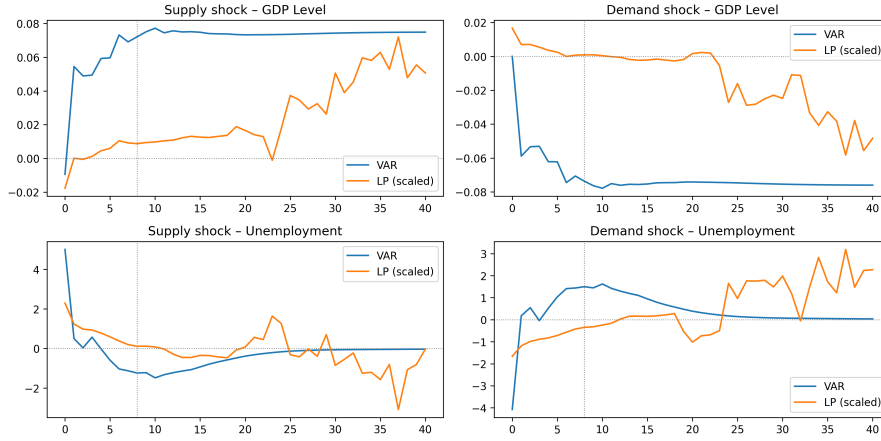


Figure 4: Only the GDP response is cumulative; Unemployment rate is in levels.

In this case, the impulse response functions (IRFs) are not particularly similar, despite all the necessary assumptions described by Wolf (2021) being taken into account. For the unemployment rate, the dynamics are roughly similar up to the horizon equal to the lag order p , but unlike the case of Christiano, Eichenbaum, and Evans (2005), the strong similarities observed there are not evident here.

5 Sign Identification: Uhlig (2005)

Uhlig (2005) adopts a sign restrictions approach to identify the effects of a monetary policy shock. The key identifying restrictions imposed on the shock's impact for the next 4 periods are:

- Prices (GDP deflator) **decrease**,
- Commodity prices **decrease**,
- Non-borrowed reserves **decrease**,
- Federal funds rate **increases**.

The GDP variable itself is left unrestricted, as it represents the primary variable of interest in the analysis.

Variables Used in Uhlig (2005)

In the original paper, the monthly variables used are:

Real GDP, GDP deflator, Commodity price index, Total reserves, Non-borrowed reserves, Federal funds rate, with a lag order of $p = 12$.

In this study, I employ quarterly data and set the lag length to $p = 4$, adapting the original approach accordingly.

While Uhlig (2005) used variables in levels, I use them in first differences.

VAR Approach

After estimating the reduced-form VAR model

$$Y_t = A_1 Y_{t-1} + \dots + A_p Y_{t-p} + u_t, \quad u_t \sim \mathcal{N}(0, \Omega),$$

we want to recover the **structural shocks**, focusing on the monetary policy shock.

The residuals u_t are linear combinations of structural shocks ε_t :

$$u_t = \tilde{P} \varepsilon_t, \quad \text{with} \quad \mathbb{E}[\varepsilon_t \varepsilon_t'] = I.$$

This implies that the residual covariance matrix satisfies

$$\Omega = \tilde{P} \tilde{P}',$$

where $\tilde{P}$ can be computed via the **Cholesky decomposition** of Ω .

Next, we want to explore different possible directions of structural shocks that are consistent with the data. To do this, we generate many random vectors $\alpha \in \mathbb{R}^m$ that have length one (i.e., $\|\alpha\| = 1$) and are mutually orthogonal (called orthonormal vectors).

This is achieved by first creating a random matrix W with entries drawn from a standard normal distribution. We then perform a QR decomposition on this matrix, which produces an orthonormal basis Q . The columns of Q serve as candidate vectors α .

Each candidate vector α defines a possible structural shock direction:

$$a = \tilde{P} \alpha,$$

where $\tilde{P}$ is the Cholesky factor of the covariance matrix of the VAR residuals.

For each candidate a , we compute the corresponding impulse responses at horizon k as a weighted sum of the standard impulse responses $r_i(k)$:

$$r_a(k) = \sum_{i=1}^m \alpha_i r_i(k),$$

where $r_i(k)$ is the response to the i -th standard Cholesky shock.

We then check whether these candidate impulse responses satisfy the **sign restrictions** over the first K periods (for example, 4 quarters):

- Prices decrease,
- Commodity price index decreases,
- Non-borrowed reserves decrease,
- Federal funds rate increases.

Only those vectors a that satisfy all these restrictions are accepted as valid structural shocks.

*Note: Since most variables are in first differences (except for the interest rate), the sign restrictions are imposed on the **cumulative** impulse responses to properly reflect their level effects.*

LP Approach

To estimate the reduced-form impulse response functions (IRFs) with respect to the forecast errors $\{u_t\}$, local projection (LP) regressions are run for each variable y_t and horizon h . Specifically, for each horizon h , the following regression is estimated:

$$y_{t+h} = \tilde{\mu}_h + \check{\beta}'_h w_t + \sum_{\ell=1}^p \check{\delta}'_{h,\ell} w_{t-\ell} + \check{\xi}_{h,t},$$

where $\check{\beta}_h \in \mathbb{R}^{n_w}$ represents the coefficient vector on the contemporaneous variables w_t . These coefficients correspond to the reduced-form IRFs of y with respect to the forecast errors u_t .

Similarly, for the policy variable r_t , the same regression is estimated to obtain coefficients $\check{\beta}_s^r$ associated with r_t at each horizon s :

$$r_{t+s} = \tilde{\mu}_s + \check{\beta}_s^{r'} w_t + \sum_{\ell=1}^p \check{\delta}_{s,\ell}^{r'} w_{t-\ell} + \check{\xi}_{s,t}^r.$$

The structural shock direction $\varepsilon_{1,t}$ is modeled as a linear combination of the forecast errors:

$$\varepsilon_{1,t} = b' u_t,$$

with $b \in \mathbb{R}^{n_w}$ a vector to be determined.

To identify b , a normalization on the immediate impact of the shock on the policy variable r_t is imposed:

$$b' \check{\beta}_0^r = 1,$$

and cumulative sign restrictions are applied on the responses of r_t for horizons $s = 1, \dots, \bar{H}$:

$$b' \sum_{s=0}^h \check{\beta}_s^r \geq 0, \quad \forall h \leq \bar{H}.$$

In other words, the vector b must ensure that the cumulative responses of r_t satisfy the imposed sign restrictions over the entire horizon of interest.

Wolf (2021) demonstrates that the identification problem thus consists in finding, for each horizon h , the vector b that maximizes (or minimizes) the structural response of y subject to the normalization and sign restriction constraints:

$$\max_{b \in \mathbb{R}^{n_w}} b' \tilde{\beta}_h \quad \text{subject to} \quad b' \tilde{\beta}_0^r = 1, \quad b' \tilde{\beta}_s^r \geq 0, \quad s = 1, \dots, \bar{H}.$$

In this way, the sign restrictions become linear constraints on the combinations of LP coefficients, and the set of structural responses consistent with the theoretical assumptions is identified.

Instead of directly solving the optimization problem, in my work a sampling method to find structural shocks that satisfy the sign restrictions is used.

1. It generates many random unit vectors b (which represent the structural shock).
2. Each vector b is adjusted so that its immediate effect on the normalization variable (i.e., interest rate) is positive, to satisfy the normalization condition.
3. For each b , it checks if the cumulative impulse responses of the restricted variables meet the sign restrictions over the given horizon (for example, prices going down, interest rate going up). If any restriction is violated, that b is discarded.
4. The vectors b that pass all these checks are accepted.
5. For each accepted b , I compute the structural impulse responses as a linear combination of the local projection coefficients at each horizon.
6. The result is a large set of possible impulse responses consistent with the imposed sign restrictions and normalization. Then the median IRF for each variable is taken.

In short, the method randomly searches for shock directions that obey the rules, keeps only those that fit, and then calculates their effects. This is an easier and flexible way to approximate the set of valid structural shocks without complex optimization.

VAR vs LP Scaled Cumulative IRF: Uhlig (2005)

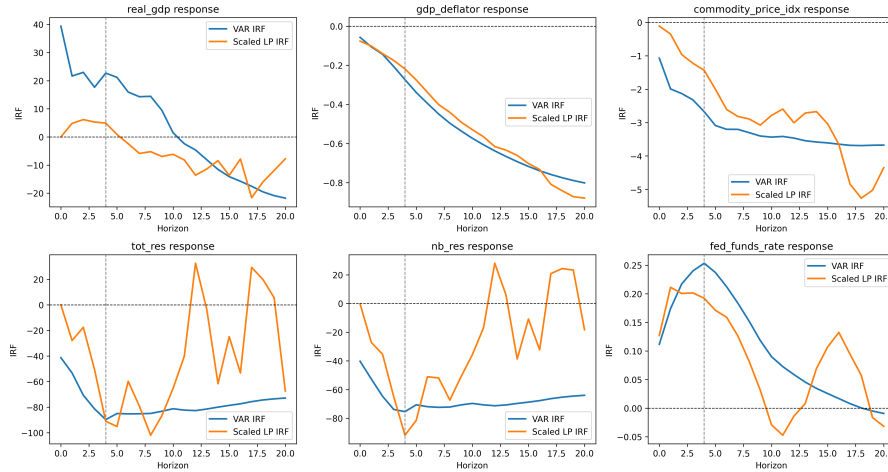


Figure 5: All the IRFs are cumulative except for the Fed Funds rate that entered the model in levels.

6 Conclusion

Wolf (2021) provides an excellent starting point for investigating a developing research topic. In my work, I attempted to follow the approach outlined in Section 3.2 of his paper, in order to implement an empirical strategy aimed at uncovering evidence supporting the theoretical results. Although the impulse response functions (IRFs) obtained through VAR and local projection (LP) methods are not always identical, I found the identification approach proposed for LPs particularly interesting. In the first case, following Christiano, Eichenbaum, and Evans (2005), the IRFs tend to be highly similar, as the LP method effectively "relies on" the shock identified by the VAR. Conversely, in the other two cases—where the shock is directly identified by the LP method itself—the IRFs generally follow similar dynamics, though they are not exactly identical.